SUVROJEET PAUL

CYBERSECURITY ENTHUSIAST | FULL-STACK & SECURE SOFTWARE DEVELOPER

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PROFESSIONAL SUMMARY

Highly motivated Computer Science & Engineering undergraduate specializing in secure systems, full-stack development, and cybersecurity research. Passionate about building scalable digital ecosystems integrating secure backend logic with intuitive user experiences.

EDUCATION

Institute of Engineering and Management (IEM), Kolkata **CGPA: 9.21**

B.Tech in Computer Science & Engineering (Networks)

Institutional Affiliation: Institute of Engineering and Management under the school of University of Engineering and Management (UEM), Kolkata.

Authpur National Model School

Secondary and Higher Secondary Education

TECHNICAL SKILLS

Programming & Tools	Python, Java, C, Git, R Studio, Excel, SQL, Linux
Cybersecurity & Hardware	Network Security, Side-Channel Analysis, Vulnerability Mapping, Secure Coding, Cryptographic Vulnerabilities
AI & Data Science	Machine Learning, Artificial Intelligence, Edge AI, Data Analysis, Data Extraction, Data Manipulation, Data Cleaning
Data Visualization	Tableau, MS Word

RESEARCH & TECHNICAL EXPERIENCE

Academic Research Assistant - Hardware & Cyber Security

- Investigating cryptographic hardware vulnerabilities using Side-Channel Power Analysis.
- Analyzing embedded systems through timing and power-consumption leakage patterns.
- Researching secure hardware architectures and low-level attack mitigation.

Open Source & Secure Full-Stack Developer

- Built scalable web and mobile applications with secure coding practices.
- Integrated backend architectures with optimized frontend interfaces.
- Applied OWASP-oriented mitigation techniques during development.

Real-Time Crypto Fraud Attribution & Blockchain Intelligence System

- Developing a blockchain intelligence system to trace suspicious cryptocurrency wallets, identify associated exchanges (VASPs), and detect fund-laundering patterns across multiple blockchain networks.
- Integrating transaction graph analysis, cross-chain tracing, and AI/ML-assisted risk detection to generate actionable intelligence for cybercrime investigations.

Hybrid GNN-RAG Framework for Cyber Threat Attribution

- Researching a hybrid Graph Neural Network (GNN) and Retrieval-Augmented Generation (RAG) architecture for multi-hop cyber-threat attribution using heterogeneous threat intelligence graphs and unstructured CTI data.
- Combining graph-based relational reasoning with evidence-grounded LLM generation to improve APT attribution accuracy, explainability, and verification of Indicators of Compromise (IOCs).

FEATURED PROJECTS

Neural Fraud Detection Engine

Developed an anomaly detection framework capable of identifying suspicious financial transactions using machine learning models and real-time behavioral analysis.

Microsoft InnerEye Architecture Implementation

Recreated architectural concepts inspired by Microsoft's InnerEye system for medical imaging analysis and ML-driven clinical support.

TimeSync: Intelligent Academic Scheduler

Designed an academic productivity platform optimizing task scheduling, resource allocation, and student workflow efficiency.

IoT Smart Safety Helmet

Built an Arduino Uno-based safety helmet integrating alcohol-detection sensors with ignition-lock mechanisms for enhanced transportation safety.

Real-Time Crypto Fraud Attribution & Blockchain Intelligence System

Developing an automated blockchain intelligence platform to trace suspicious cryptocurrency wallets, identify. associated Virtual Asset Service Providers (VASPs), and analyze cross-chain fund movements using transaction graph analysis and AI/ML-assisted risk detection.

Hybrid GNN-RAG Framework for Cyber Threat Attribution

Designing a hybrid Graph Neural Network (GNN) and Retrieval-Augmented Generation (RAG) framework integrating threat knowledge graphs with contextual cyber threat intelligence to improve multi-hop APT attribution, IOC correlation, explainability, and evidence-based threat analysis.

PROFESSIONAL CERTIFICATIONS

- Programming, Data Structures and Algorithms using C — NPTEL / CMI
- Discrete Mathematics — IIT Madras
- Foundations of Project Management — Google
- Information Theory — CUHK
- Cyber Security Fundamentals — University of London
- Advanced Systems Security Topics — University of London
- Classical cryptosystem and core concepts – University of Colorado
- Mathemetical concepts for cryptography– University of Colorado
- Advanced course on cybersecurity and data analytics- Asian Institute of technology
- Cybersecurity and Privacy – IIT Madras

LEADERSHIP **& COMMUNITY ENGAGEMENT**

IEEE Microwave Theory and Technology Society (MTT-S) • IEEE Computer Society • Google Gemini Student Community (UEMK Chapter) • GeeksforGeeks Student Chapter • IEEE Communications Society

LANGUAGES

English (Professional Proficiency) • Hindi (Professional Proficiency) • Bengali (Native/Bilingual)